

Field Theory

Setup

Exercise 1. Give an example of an infinite field whose characteristic is not 0.

Exercise 2.

- (a) Suppose F_1, F_2 are fields and $\phi: F_1 \rightarrow F_2$ is a ring homomorphism. Show that ϕ is either injective or is constantly equal to 0.
- (b) Suppose F is a field of characteristic p . Prove that F has a subfield isomorphic to $\mathbb{Q}$ if $p = 0$, or to $\mathbb{F}_p$ if $p > 0$.
- (c) Find an example of a field F and an injective homomorphism $\phi: F \rightarrow F$ which is not surjective.

Exercise 3. If a field K is an extension of a field F , then K can also be thought of as a vector space over F .

- (a) Consider $\mathbb{Q}(\sqrt{2})$ as a $\mathbb{Q}$ -vector space. Find a basis.
- (b) Consider $\mathbb{Q}(\sqrt[3]{2})$ as a $\mathbb{Q}$ -vector space. Find a basis.
- (c) Suppose $f \in \mathbb{Q}[x]$ is irreducible of degree n , and consider $\mathbb{Q}[x]/(f)$ as a $\mathbb{Q}$ -vector space. Find a basis.

Exercise 4. If K is an extension of F , its **degree** $[K : F]$ is the dimension of K as an F -vector space.

- (a) Suppose $L \supset K \supset F$ is a tower of extensions. Prove that $[L : F] = [L : K][K : F]$ (that is, degrees multiply in towers).
- (b) Suppose $f \in \mathbb{Q}[x]$ is irreducible, and $\alpha \in \mathbb{C}$ is a root of f . What is $[\mathbb{Q}(\alpha) : \mathbb{Q}]$?
- (c) Give examples of distinct elements $\alpha, \beta \in \mathbb{C}$ such that

$$[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}] < [\mathbb{Q}(\alpha) : \mathbb{Q}][\mathbb{Q}(\beta) : \mathbb{Q}].$$

Can you find an example where $\alpha \notin \mathbb{Q}(\beta)$ and $\beta \notin \mathbb{Q}(\alpha)$?

Exercise 5. A polynomial $f \in \mathbb{Q}[x]$ is said to be **separable** if it has distinct roots in $\mathbb{C}$, and **inseparable** otherwise.

- (a) Give an example of $f \in \mathbb{Q}[x]$ which is inseparable and reducible.
- (b) Prove that f is separable if and only if $\gcd(f, f') = 1$.
- (c) Is it possible for $f \in \mathbb{Q}[x]$ to be inseparable *and* irreducible?

Exercise 6. Here are two facts you should know:

- If K is a field of characteristic p , then for any $a, b \in K$, $(a + b)^p = a^p + b^p$ by the Binomial Theorem. This means the function $\text{Fr}: K \rightarrow K$ defined by $\text{Fr}(x) = x^p$ is a ring homomorphism. (And it is always injective by Exercise 2.)
 - (Fermat's Little Theorem) In $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, every element satisfies $x^p - x = 0$. So, $\text{Fr} = \text{id}$, and in particular is surjective.
- (a) Suppose K is a finite field of characteristic p . Prove that Fr is surjective, hence is an isomorphism $K \rightarrow K$.
- (b) Let $K = \mathbb{F}_3(i)$, where i is a root of the polynomial $x^2 + 1$ (which is irreducible over $\mathbb{F}_3$). Check that, in this case, $\text{Fr} \neq \text{id}$.

Punchline

Exercise 7. (August 2023 Problem 2) Let p be an odd prime number. Prove that any two fields of order p^2 are isomorphic. (Do not just quote the theorem that there's a unique isomorphism class of fields of any prime power order; your job is to prove a case of that theorem!)

Exercise 8. (January 2025 Problem 3) Let F be a field, $K \supset F$ be a finite extension of F , and let a be an element of K .

- (a) Suppose the degree $[K : F]$ is relatively prime to 6. Show that $F(a) = F(a^3)$.
- (b) Suppose the degree $[K : F]$ is not divisible by 3. Either prove that we still have $F(a) = F(a^3)$, or show by example that this is false.

Exercise 9. (August 2022 Problem 4) Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$).

- (a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.
- (b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.
- (c) Show that K has a single degree p inseparable extension.

Exercise 10. (January 2023 Problem 4) Consider the finite field $\mathbb{F}_p$ where p is a prime number.

- (a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p .)
- (b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)